Submitted By:

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Assignment:

Pac-Man Game Report

Submitted To:

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For the fulfillment of,

Pac-Man Game

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Terminal-Based Pacman Game (Java OOP)

Introduction

This project is a simple terminal-based Pacman game developed in Java using Object-Oriented Programming (OOP). The player controls Pacman using keyboard inputs (<, >, ^, v). The goal of the game is to eat food, avoid the ghost, and move around a board that contains walls and obstacles.

Objectives

The main objective of this project is to build a playable Pacman game in the terminal while practicing OOP concepts such as classes, objects, and modular design. The system is designed to be simple, structured, and easy to understand.

System Overview

The game is built using multiple classes that work together. The PacmanGame class acts as the main controller and manages the game loop, user input, and overall flow of the game. The GameBoard class represents the game map and is responsible for creating the grid, placing walls and food, and displaying the board.

The Pacman class represents the player character, handling movement and score updates when food is collected. The Ghost class represents the enemy and moves toward Pacman while avoiding walls using simple logic. The Food class represents the items that give points when collected.

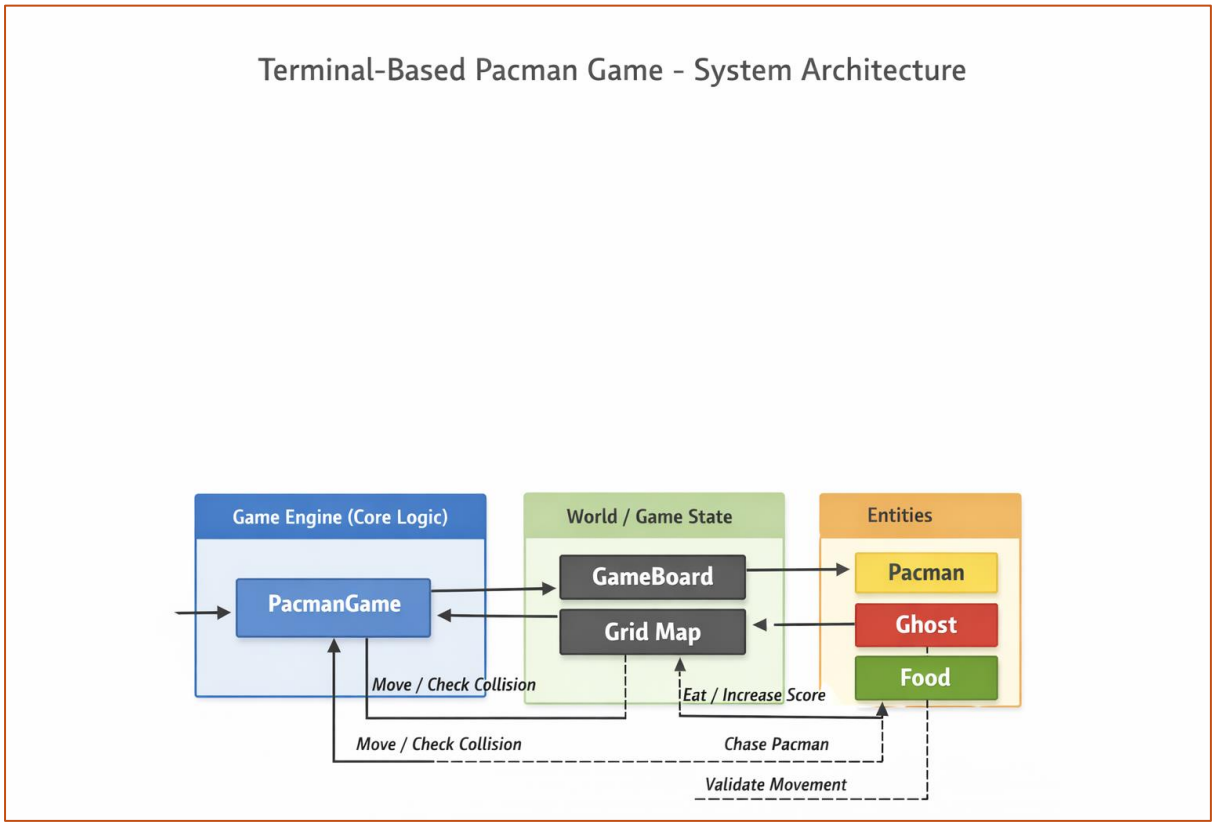
Game Flow

The game starts by initializing all objects, including the board, Pacman, and Ghost. The board is displayed in the terminal, and the player is asked to enter movement commands using keyboard inputs. Pacman moves according to the input and collects food when possible. After each move, the ghost also moves toward Pacman. The game continues in a loop until either the player quits or the ghost catches Pacman.

Features

The game runs in the terminal and includes a grid-based map with walls and obstacles. It has a score system that increases when Pacman eats food. A ghost enemy follows Pacman, creating challenge and gameplay. The design is simple and based on OOP principles.

System Architecture



OOP Concepts Used

This project uses key OOP concepts. Encapsulation is used by keeping data inside classes and accessing it through methods. Abstraction is used to hide complex logic behind simple methods. Modularity is achieved by separating the game into different classes, each with a specific responsibility. The classes interact with each other to make the game work as a whole system.

Limitations

The game uses a basic ghost AI that does not include advanced pathfinding. There is only one ghost in the game, and there are no multiple levels or difficulty settings. The game also does not include a graphical interface.

Future Improvements

Future improvements can include adding more ghosts with different behaviors, implementing pathfinding algorithms, introducing multiple levels, and improving the visual display in the terminal.

Conclusion

This project shows how a simple game like Pacman can be built using Java and OOP principles. By dividing the system into multiple classes, the code becomes easier to manage and extend. It is a good example of how object-oriented design can be used in game development.